Google Cloud

Course 4 resources and citations

Module 1: Introduction to the power of storytelling: How to visualize data in the cloud

Resources

Helpful resources and tips

- Common problems with labs/lab troubleshooting
- Google Docs Help Center
- Google Sheets Help Center
- How to use Google Slides

Why visualization matters

- Information is beautiful
- Making events accessible
- Why data visualization is important
- Why is data visualization important? What is important in data visualization?

Foundational concepts of design: Structure and aesthetics

- Color choices in data visualizations: Importance and how to
- Visual hierarchy, Gutenberg diagram, F & Z pattern

Foundational concepts of design: Usability and accessibility

- Accessible data visualizations
- Data visualizations, charts, and graphs

Citations

Why visualization matters

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Ways that cloud data enhances storytelling

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The importance of UX/UI design

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Design decisions for data visualizations

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Foundational concepts of design: Structure and aesthetics

- Design, Y. (2019, May 16). Visual hierarchy, Gutenberg diagram, F & Z pattern. *Medium*.
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Foundational concepts of design: Usability and accessibility

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Design principles and strategies for dashboards

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Module 2: Methods for visualization planning and design

Resources

Visualization techniques for different data types

- Data visualization resources: Types of charts and graphs for data viz
- How to choose the best chart or graph for your data

Choose an effective visualization

- Explore new ways to visualize your data

[Supplemental] Tips for effective teamwork

- Teams start with human connections

The importance of the user journey

- Web Content Accessibility guidelines

Citations

Data types to consider for visualizations

- Dijk, E. van. (2015, October 6). Which chart type works best for your data? *Medium*.
- Google Docs Editors Help. (n.d.). *Histogram charts*. Retrieved January 4, 2024.
- Mimi. (2020, February 10). Data-heavy applications: How to design perfect charts. *Medium*.
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The importance of data type identification

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SMART questions for stakeholders

- Chang, J. (2022, March 3). Using the SMART framework to ask the effective questions. *Medium*.
- Gautam, A. (2021, September 6). Methodology for crafting effective questions for data analysis. *CodeX*.
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Strategies for translating stakeholder requests

- Haidermota, M. (2023, November 15). Stakeholder communication in analytics—It matters! *Medium*.
- Ogunkoya, O. (2023, December 6). *Translating and managing stakeholder needs in business analysis*. Linkedin.

Introduction to wireframes for stakeholder alignment

- Bahmani, H. (2023, September 19). *5 ways to build a better dashboard wireframe*. Built In.
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Module 3: Access, explore, and report on data in the cloud

Resources

Cloud-based tools for accessing and visualizing data

- Looker for business intelligence
- Data Studio

Actionable insights through data exploration

- Unlock the power of data analytics with Looker: Rethinking business intelligence

Customize dimensions and measures in Data Studio

- About the "How to" guide

Case study: Benefits of using data reports

- 11 data visualization techniques for every use-case with examples
- 17 data visualization techniques all professionals should know
- Power BI dashboards vs reports: A comprehensive guide

Citations

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Aggregate and filter to summarize data

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Data blending for helpful insights

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Introduction to data reports

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Module 4: Enterprise business analytics

Resources

Data governance for safeguarding data

- Best practices In data governance
- What is data governance?

Insights from a single dataset

- What is exploratory data analysis?

Guide to Looker Enterprise

- Create and edit dashboards and reports
- How Looker generates SQL
- Looker introduction

Citations

Introduction to enterprise-grade visualization tools

- Google Cloud. (n.d.). *Looker*. Retrieved May 23, 2023.
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Data governance for safeguarding data

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Guide to Looker Enterprise

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- Google Cloud. (2023, May 30). *Send and share content*.
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The changing role of a cloud data analyst

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Module 5: Explore the developer environment

Resources

Guide to LookML

- Introduction to LookML
- LookML data tests: Recommendations and best practices
- LookML terms and concepts
- Types of files in a LookML project
- Using version control and deploying

Explore a dashboard file's LookML code

- Dashboard element list
- LookML dashboard parameters

Citations

Bring data tools together with an IDE

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The elements of LookML

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The benefits of using version control

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Introduction to data modeling languages

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Define dimensions and measure with LookML

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Discover dashboards as code

- Google Cloud. (2023, November 17). *Building LookML dashboards*.

Explore a dashboard file's LookML code

- Google Cloud. (2023, February 1). *Dashboard element list*.

Derived tables for complex data problems

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Improve performance with caching

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